Quaternions

$$q = (a, b, c, d)$$

$$= a + b\hat{i} + c\hat{j} + d\hat{k}$$

$$= a + \vec{v}, \quad \vec{v} = \begin{bmatrix} b \\ c \\ d \end{bmatrix}$$

a: Scalar; $\vec{v}$: 3-D vector

Different representations of a quaternion, allows us to use vector algebra principles

Quaternion Rotation

$$\vec{e}' = q\,\vec{e}\,q^*$$

- $\vec{e}'$: rotated vector $= \begin{bmatrix} e_1' \\ e_2' \\ e_3' \end{bmatrix}$
- q: $(a+\vec{v})$
- q^*: Conjugate of q, Also the inverse $= (a-\vec{v})$
- $\vec{e}$: Vector to be rotated $= \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = (0+\vec{e})$

$q\vec{e}q^* =$ vector!

Scalar components $= 0$

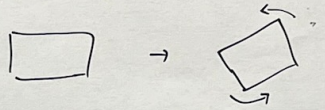
$a(0) + -\,\vec{v}\cdot(-\vec{v}) = 0$

Matrix (Euler) Rotation

2D: $\vec{e}' = R\,\vec{e}$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- Follows right hand rule.



Multiplying Quaternions

$$q\,p \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= (a+\vec{v})(w+\vec{u})$$

$$= aw + \vec{v}w + a\vec{u} + \vec{v}\vec{u}$$

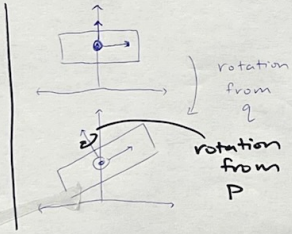
[scalar][scalar] = [scalar]; [vector][scalar] = [vector]; [scalar][vector] = [vector]; [vector][vector] = [?]

$$\vec{v}\vec{u} = -\underbrace{\vec{v}\cdot\vec{u}}_{\text{scalar}} + \underbrace{\vec{u}\times\vec{v}}_{\text{vector}}$$

This means we can stack up rotations really easily.

+ each rotation is relative to itself

$$p\,\underbrace{q\vec{e}q^*}_{\text{vector to be rotated}}\,p^*$$



3D: $\vec{e}' = R\,\vec{e}$

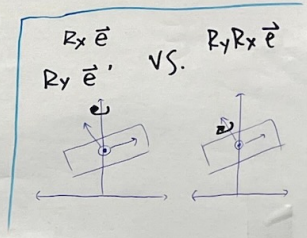
R: product of all rotations

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

usually $R = R_z R_y R_x$ (starts here)



$$(a+\vec{v})(w+\vec{u})$$

$$= \underbrace{aw - \vec{v}\cdot\vec{u}}_{\text{scalar}} + \underbrace{w\vec{v} + a\vec{u} + \vec{u}\times\vec{v}}_{\text{vector}}$$

With Quaternion multiplication, we can look into Rotations!

Rotation Quaternion

$$p = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\vec{n}$$

- unit quaternion represents a rotation by θ around $\vec{n} = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$

PROBLEM: Gimbal Lock

$$R = R_z\,R_y\,R_x$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\eta & -\sin\eta & 0 \\ \sin\eta & \cos\eta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ \sin(\theta+\eta) & \cos(\theta+\eta) & 0 \\ -\cos(\theta+\eta) & \sin(\theta+\eta) & 0 \end{bmatrix}$$

lost ϕ !!!